SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-015: Student Records Export to CSV

Change Request ID	CR-015
Change Request Title	Student Records Export to CSV
Student Name	Y.Ugatharsini
Student ID	JF/ICT/24/33
Implementation Status	Completed
Date	15 October 2026

1. Change Request Summary

1.1 Business Requirement

The Student Management System required a facility that allows administrators to export student records into a CSV (Comma-Separated Values) file for reporting, record-keeping, and external data processing.

Previously, administrators could view, search, add, edit, and delete student records through the Student Management System interface. However, a dedicated CSV export facility was required to allow administrators to download student information for administrative purposes.

The new feature introduces an Export CSV button to the administrator's student management dashboard. When the administrator clicks this button, the system generates and downloads a file named students.csv.

The exported file contains the following information:

- Student ID
- Registration Number
- First Name
- Last Name
- Email
- Phone Number

The implementation uses the student records already retrieved from the backend API. Therefore, no additional database tables, database columns, or backend API endpoints are required.

The main objective is to improve administrative efficiency and provide a convenient way to maintain and process student records while preserving existing authentication, authorization, and student management functionality.

1.2 Existing Behaviour

- Administrators could log in using their existing credentials.
- Administrators could view student records on the dashboard.
- The system displayed student registration numbers, names, email addresses, phone numbers, and last login information where available.
- Administrators could search student records using the existing search functionality.
- Administrators could add new student records.
- Administrators could edit existing student records.
- Administrators could delete student records.
- Student data was retrieved through the existing GET /api/students/ endpoint.
- The backend restricted access to student management operations using authentication and role-based authorization.
- A dedicated facility for downloading student records as a CSV file was not available.

1.3 Requested Change

The requested change was to add an Export CSV button to the administrator's student management dashboard. When selected, the feature should create and download a properly formatted students.csv file containing the required student fields and column headers.

The export should support multiple student records, handle an empty list without causing an application error, and preserve existing student viewing, searching, adding, editing, and deleting functionality. Access to student records must continue to be protected by the existing administrator authentication and authorization controls.

The provided implementation description states that the feature uses records already retrieved from the backend API, so it does not require a new database schema or backend API endpoint.

1.4 Implementation Approach

The frontend uses the student records already available in the dashboard and converts the relevant fields into CSV rows. The first row contains the column headers. Each subsequent row represents one student. The generated CSV is downloaded to the administrator's device with the filename students.csv.

To maintain CSV formatting, values containing commas, quotation marks, or line breaks should be escaped correctly. The export action should not update, delete, or otherwise modify the original student records.

2. Expected CSV Structure

Student ID	Registration Number	First Name	Last Name	Email	Phone Number
STU001	REG001	John	Perera	john@example.com	0771234567

Illustrative example only: the actual exported rows should reflect the student records available in the system.

3. Security and Data Integrity

- Only authenticated administrators should be able to access the export control and the student records it exports.
- The existing authentication and role-based authorization controls must remain in place.
- The export operation must not modify or delete the original student records.
- The export must accurately reflect the student data available to the administrator.

4. Testing Checklist

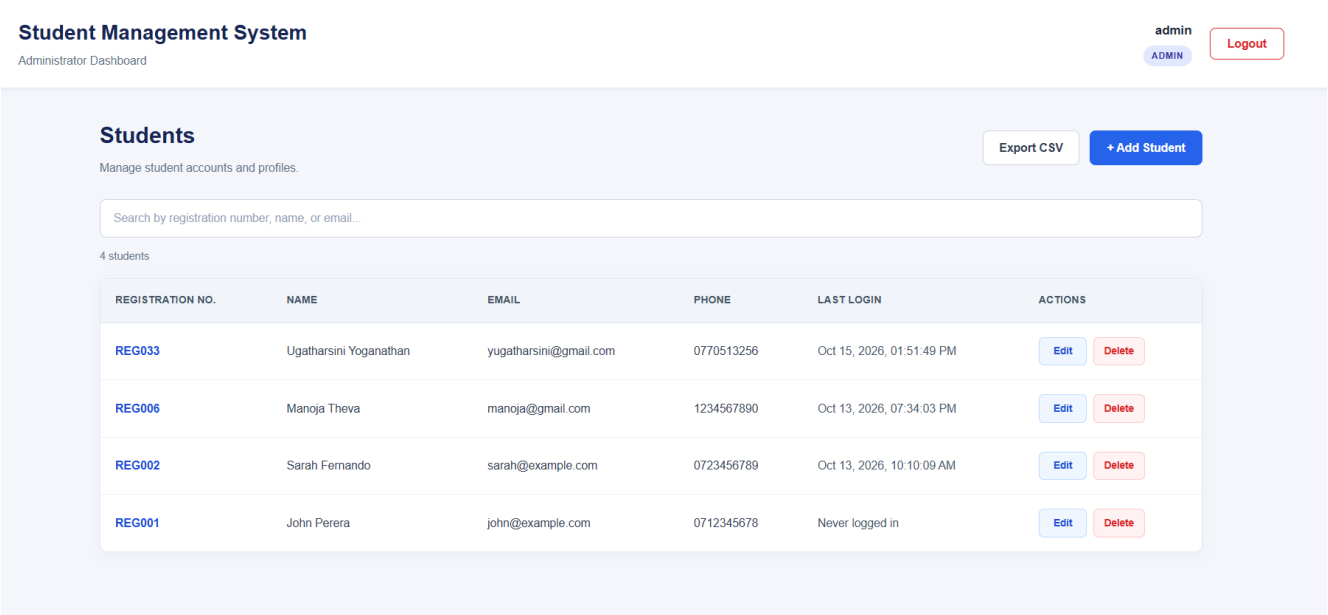
Test ID	Test Description	Result
TC-01	Export CSV button is visible to an administrator.	To be verified
TC-02	Clicking Export CSV downloads students.csv.	To be verified
TC-03	CSV contains all six required column headers.	To be verified
TC-04	Exported student values match the dashboard records.	To be verified
TC-05	Multiple student records are exported correctly.	To be verified
TC-06	An empty student list does not cause a runtime error.	To be verified
TC-07	Unauthorized users cannot access student export functionality.	To be verified
TC-08	Existing view, add, edit, and delete functions continue to work.	To be verified
TC-09	Exporting does not modify or delete original student records.	To be verified

5. Acceptance Criteria Verification

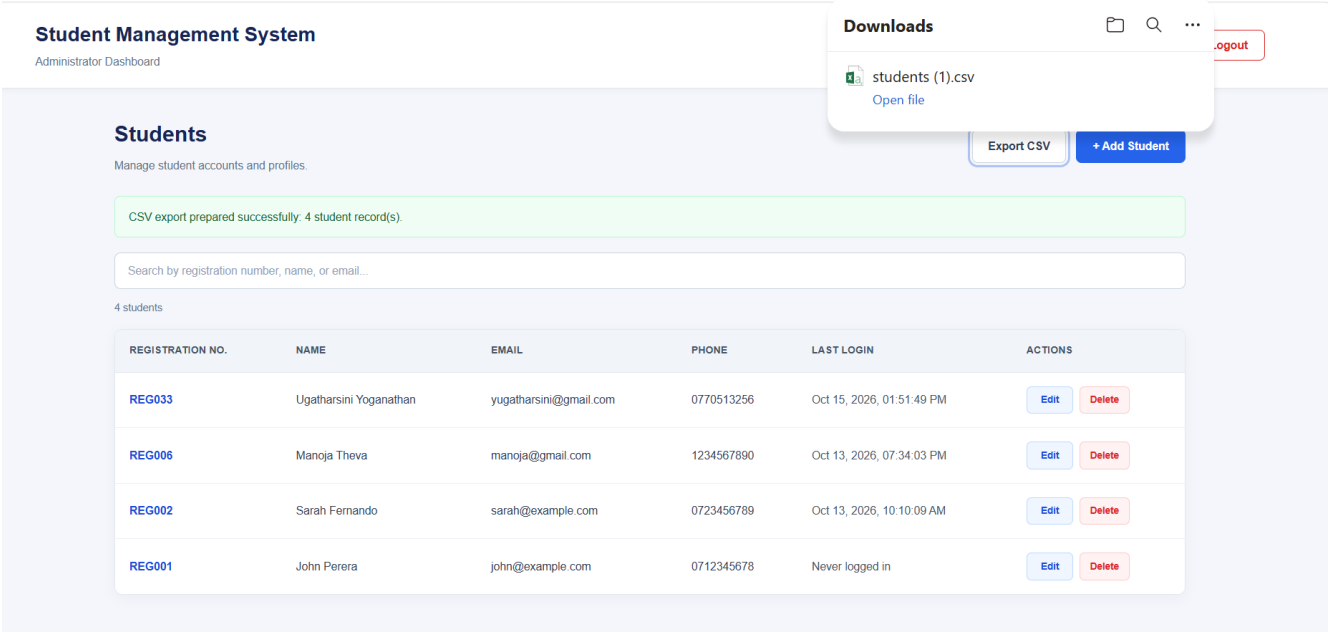
The change can be marked fully verified after the tests above have been executed and the evidence has been captured. The implementation status supplied in the report header is “Completed”; test results should be updated to reflect the actual outcomes before final submission.

6. Evidence to Include

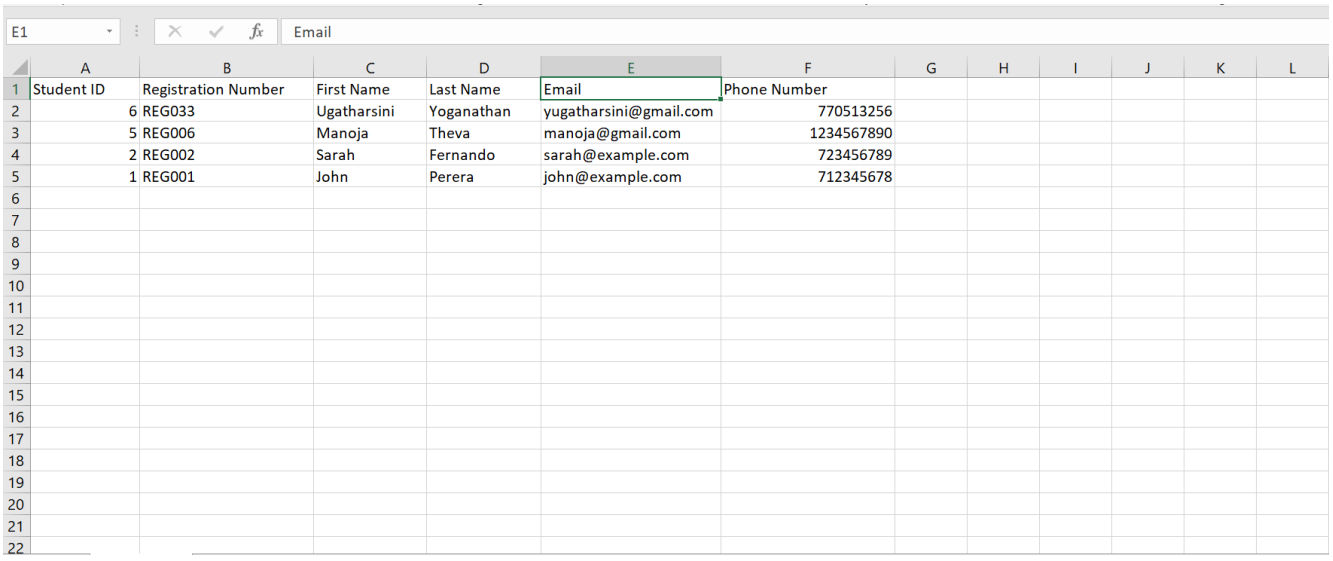
Evidence 01 – Screenshot of the administrator student management page showing the Export CSV button.



Evidence 02 – Screenshot or browser evidence showing that students.csv was downloaded successfully.



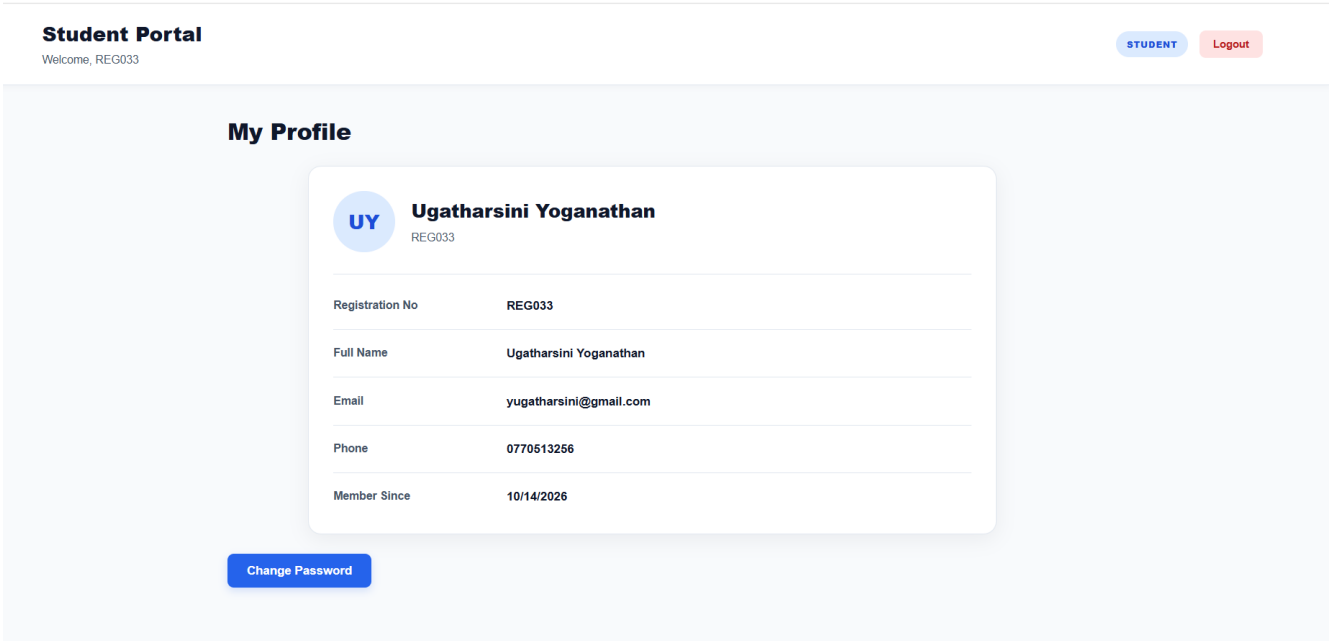
Evidence 03 – Screenshot of students.csv opened in spreadsheet software, showing the required headers and records.



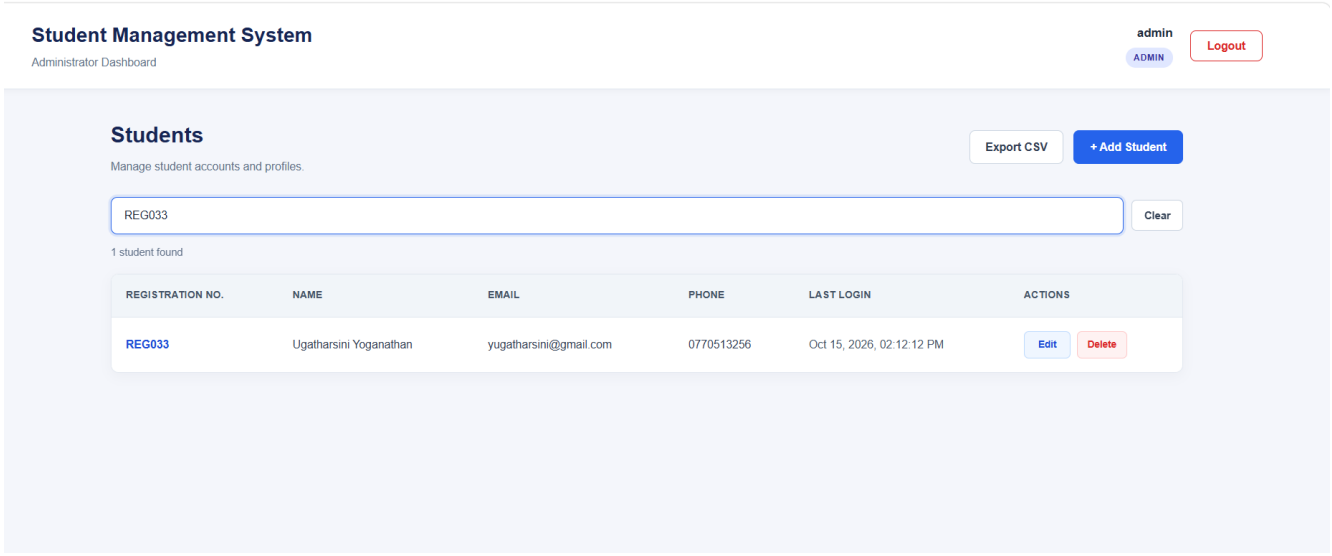
E1 | Email

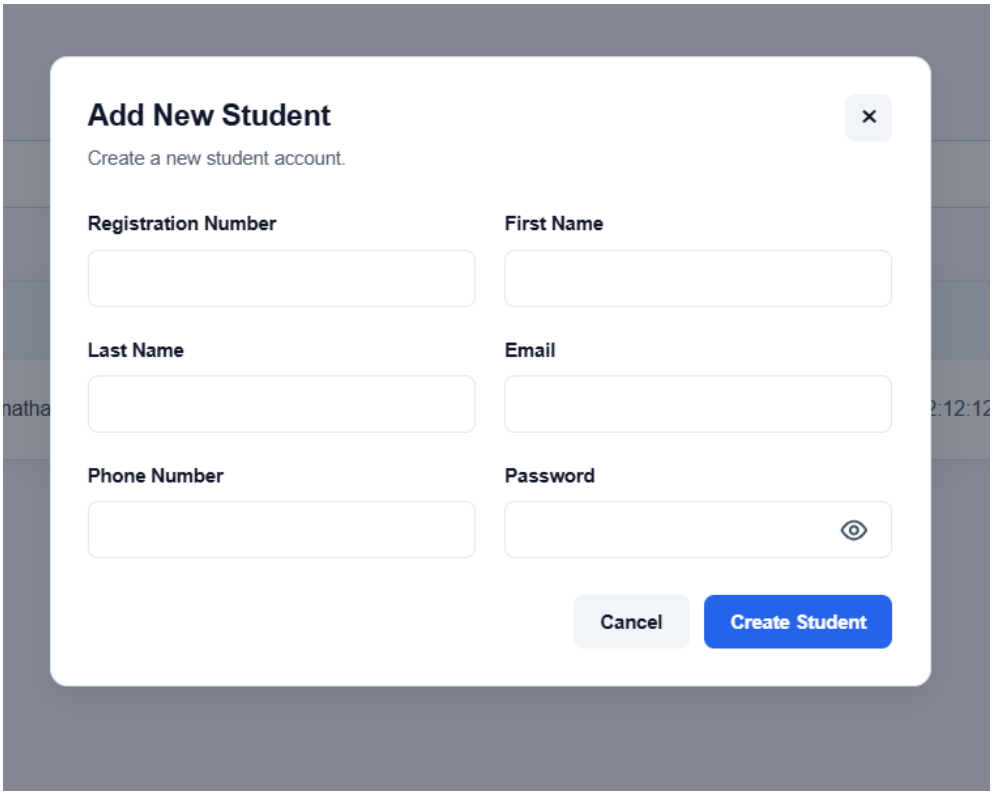
	A	B	C	D	E	F	G	H	I	J	K	L
1	Student ID	Registration Number	First Name	Last Name	Email	Phone Number						
2	6	REG033	Ugatharsini	Yoganathan	yugatharsini@gmail.com	770513256						
3	5	REG006	Manoja	Theva	manoja@gmail.com	1234567890						
4	2	REG002	Sarah	Fernando	sarah@example.com	723456789						
5	1	REG001	John	Perera	john@example.com	712345678						
6												
7												
8												
9												
10												
11												
12												
13												
14												
15												
16												
17												
18												
19												
20												
21												
22												

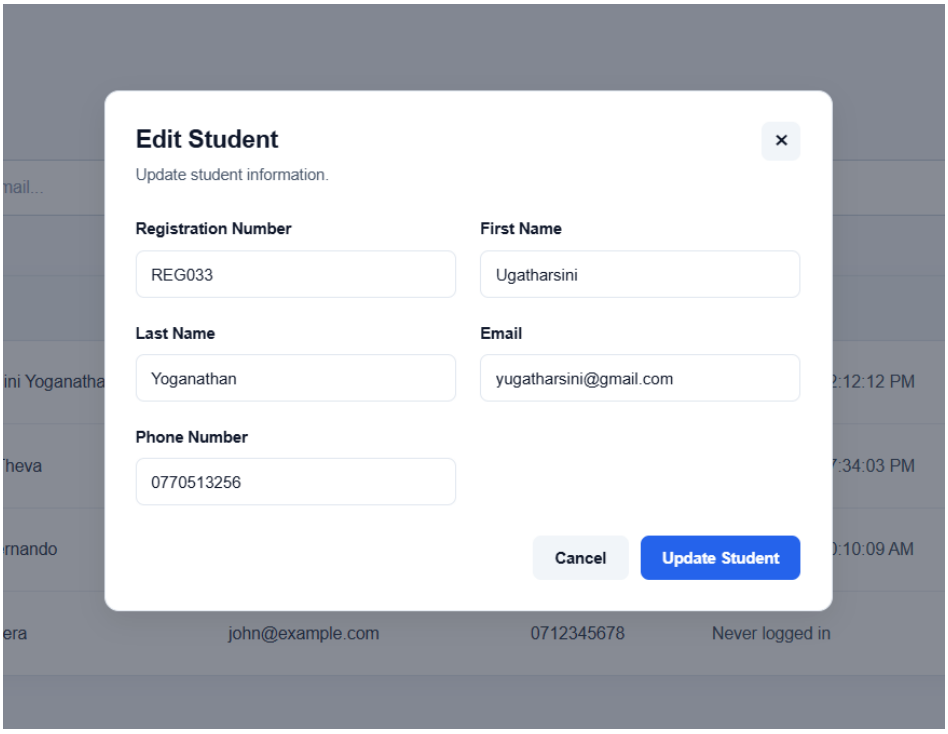
Evidence 04 – unauthorized users cannot export student records, where applicable.

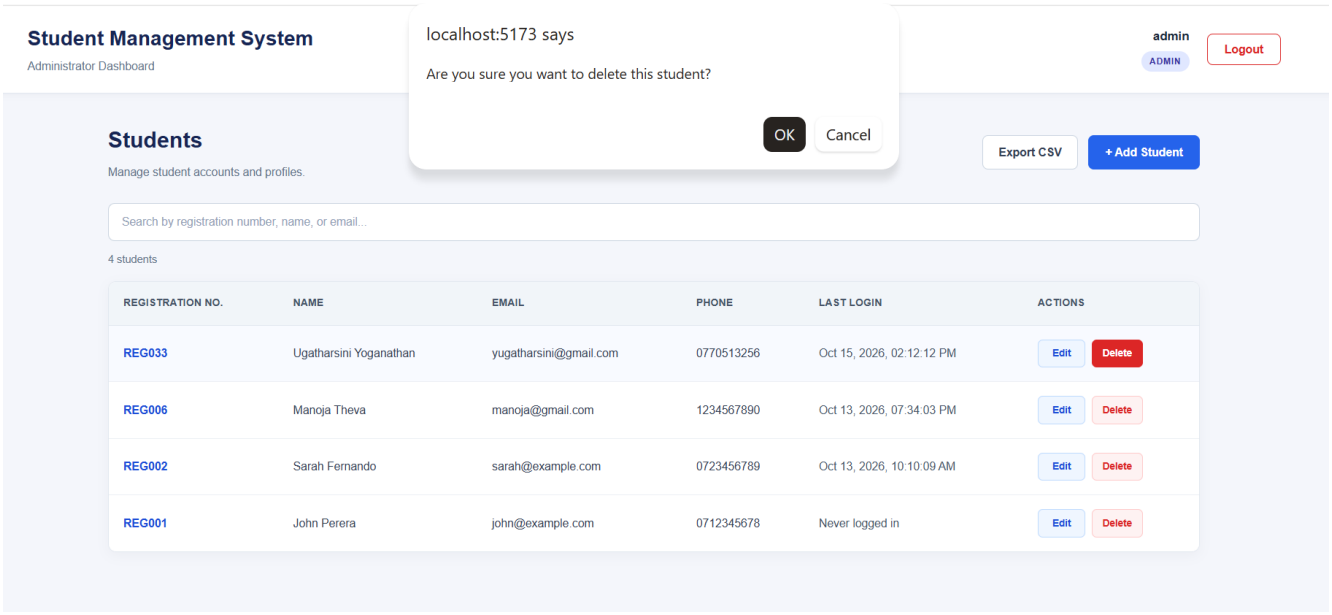


Evidence 05- existing student management functions continue to work.









7.1 Final Summary

CR-015 was implemented to provide a Student Records Export to CSV feature for the Student Management System.

The implementation enables administrators to download student records in a CSV file named students.csv directly from the administrator dashboard. The exported file contains Student ID, Registration Number, First Name, Last Name, Email, and Phone Number, allowing student information to be used for reporting, record-keeping, and external data processing.

The frontend implementation generates the CSV file using the student records retrieved from the existing backend API. The CSV generation process includes meaningful column headers, appropriate

field escaping, handling of empty values, and support for special characters such as commas and quotation marks. A UTF-8 byte-order mark is included to improve compatibility with spreadsheet applications.

The Export CSV button also provides appropriate behaviour during data loading and when student data retrieval fails. The dashboard stylesheet was updated to provide consistent button styling, hover effects, disabled states, and responsive layout behaviour.

The existing getStudents() controller, getAllStudents() model function, and administrator-only student records endpoint have been reused without introducing a new backend API endpoint or modifying the database schema. Existing authentication and role-based authorization mechanisms have been preserved.

Functional, CSV formatting, data validation, error-handling, responsive interface, authorization, and regression testing were conducted to verify the export functionality and ensure that existing student search, Add Student, Edit Student, and Delete Student operations continue to work correctly.

The implementation is considered complete when CSV generation, downloaded file contents, authorization, and existing student management functionality have been verified successfully. The change improves administrative efficiency while preserving the existing system functionality and student records.

7.2 Issues / Limitations

The following configuration and operational considerations apply:

- The CSV export feature depends on the successful retrieval of student records from the existing backend API.
- The export button remains disabled while student records are loading or when the data request has failed.
- If no student records are available, the system displays an appropriate message instead of generating a CSV file containing student records.
- The export includes the student records currently available in the frontend state. If server-side pagination is introduced in the future, additional handling may be required to export all records across multiple pages.
- The exported CSV file contains student personal information and must be stored and shared securely.
- CSV escaping preserves field boundaries when values contain commas or quotation marks. Formula injection protection must also be maintained and tested to reduce risks when CSV files are opened in spreadsheet applications.
- The implementation supports CSV export only. PDF and Excel export formats are outside the scope of CR-015.
- Existing authentication and administrator authorization must remain enabled to prevent unauthorized access to student records.
- The export operation does not modify or delete student information in the database.
- No database schema changes or additional backend API endpoints are required.

Outstanding Issues: None identified, provided that CSV generation, authorization, data accuracy, and regression tests have been verified successfully.

Student Name: Y.Ugatharsini

Student ID: JF/ICT/24/33

Date: 15 October 2026